

L7 Planning Quiz ANS

1. What is the primary output of mission planning in an autonomous-driving system?

- A) A map-level route from the start to the destination
 - B) A steering-angle command
 - C) A set of detected objects
 - D) A throttle and brake signal
- ANS: A) A map-level route from the start to the destination

2. What is the primary output of behavior planning?

- A) A raw occupancy grid
 - B) A selected high-level driving maneuver, such as stop, follow, or change lanes
 - C) A sequence of actuator voltages
 - D) A global HD map
- ANS: B) A selected high-level driving maneuver, such as stop, follow, or change lanes

3. What does motion planning generate for the vehicle controller to track?

- A) A traffic-rule database
 - B) A destination address
 - C) A feasible reference trajectory
 - D) A sensor-calibration table
- ANS: C) A feasible reference trajectory

4. In route planning, the road network is commonly represented as which structure?

- A) An unweighted image
 - B) A finite-state controller
 - C) A speed profile
 - D) A weighted graph
- ANS: D) A weighted graph

5. What problem does route planning solve?

- A) Finding an optimal navigation path from start to goal
 - B) Predicting the color of a traffic light
 - C) Estimating tire pressure
 - D) Selecting a camera exposure
- ANS: A) Finding an optimal navigation path from start to goal

6. Which value does Dijkstra's algorithm use to prioritize a node?

- A) Only the estimated cost from that node to the goal
 - B) The currently known cost from the start to that node
 - C) The node's geographic latitude
 - D) The number of outgoing edges only
- ANS: B) The currently known cost from the start to that node

7. In A* search, what does the evaluation value normally combine?

- A) Vehicle speed and steering angle
 - B) Road width and lane count
 - C) Cost from the start and estimated cost to the goal
 - D) Camera and LiDAR confidence
- ANS: C) Cost from the start and estimated cost to the goal

8. What property must a heuristic have to be admissible for A* optimality?

- A) It must always equal zero
 - B) It must overestimate the remaining cost
 - C) It must ignore the goal
 - D) It must never overestimate the true remaining cost
- ANS: D) It must never overestimate the true remaining cost

9. Why can A* explore fewer positions than Dijkstra's algorithm?

- A) Its heuristic directs the search toward the target
- B) It ignores all edge costs
- C) It expands nodes randomly
- D) It searches only backward from the goal

ANS: A) Its heuristic directs the search toward the target

10. Which information should a behavior planner consider when choosing a driving maneuver?

- A) Only the destination coordinates
- B) Road rules plus static and dynamic objects
- C) Only the current steering angle
- D) Only the vehicle's battery level

ANS: B) Road rules plus static and dynamic objects

11. What does the 'Follow Leader' maneuver require the ego vehicle to do?

- A) Pass the leading vehicle immediately
- B) Stop regardless of traffic
- C) Match the leader's speed while maintaining a safe distance
- D) Ignore the leading vehicle

ANS: C) Match the leader's speed while maintaining a safe distance

12. In a finite-state-machine behavior planner, what normally triggers a transition between states?

- A) A randomly selected timer
- B) A software recompilation
- C) A change in map file format
- D) An input condition or event

ANS: D) An input condition or event

13. How does reinforcement-learning-based behavior planning learn a policy?

- A) Through trial and error to maximize cumulative reward
- B) By storing only hand-written traffic rules
- C) By applying Dijkstra's algorithm to every maneuver
- D) By eliminating all environmental feedback

ANS: A) Through trial and error to maximize cumulative reward

14. Why might an FSM use $Ego.Velocity \leq StopThreshold$ instead of $Ego.Velocity == 0$?

- A) To increase the maximum speed
- B) To improve robustness to state-estimation noise
- C) To disable stopping behavior
- D) To remove the need for sensors

ANS: B) To improve robustness to state-estimation noise

15. Why is there no direct transition from 'Decelerate to Stop' to 'Track Speed' at a stop-sign intersection?

- A) The vehicle must always turn left
- B) The intersection has no approaching zone
- C) The vehicle must first come to a complete stop
- D) The behavior planner cannot measure speed

ANS: C) The vehicle must first come to a complete stop

16. How does a hierarchical FSM organize multiple driving scenarios?

- A) It uses one state for all situations
- B) It removes all entry and exit transitions
- C) It replaces behavior planning with route planning
- D) It models each scenario as a super-state containing a scenario-specific lower-level FSM

ANS: D) It models each scenario as a super-state containing a scenario-specific lower-level FSM

17. Which is a major limitation of FSM-based behavior planning?

- A) State explosion as scenarios become numerous and complex

- B) Inability to represent any driving maneuver
- C) Requirement that every road be straight
- D) Inability to use logical conditions

ANS: A) State explosion as scenarios become numerous and complex

18. How is a trajectory distinguished from a geometric path?

- A) A trajectory contains no spatial information
- B) A trajectory is a path parameterized by time
- C) A trajectory is always a straight line
- D) A trajectory contains only map-level turns

ANS: B) A trajectory is a path parameterized by time

19. Into which two main tasks can motion planning be decomposed?

- A) Localization and mapping
- B) Detection and segmentation
- C) Path planning and speed-profile generation
- D) Mission planning and sensor fusion

ANS: C) Path planning and speed-profile generation

20. What does an occupancy grid encode?

- A) Only the vehicle's control gains
- B) Only the destination and start points
- C) Only dynamic-object identities
- D) Occupied and free regions in a discretized environment

ANS: D) Occupied and free regions in a discretized environment

21. What is a key difference between RRT and PRM when start and goal locations change?

- A) RRT typically grows a new graph, whereas PRM can reuse its roadmap
- B) PRM cannot represent obstacles
- C) RRT always reuses one fixed roadmap
- D) PRM does not require path search

ANS: A) RRT typically grows a new graph, whereas PRM can reuse its roadmap

22. After constructing a probabilistic roadmap, which algorithms can be used to find a route through it?

- A) PID or MPC
- B) A* or Dijkstra's algorithm
- C) K-means or PCA
- D) Backpropagation or gradient descent

ANS: B) A* or Dijkstra's algorithm

23. What does it mean that many RRT and PRM variants are probabilistically complete?

- A) They always return the shortest path
- B) Their runtime is fixed and predictable
- C) If a path exists, it will be found in finite time, although the time may be very long
- D) They never require collision checking

ANS: C) If a path exists, it will be found in finite time, although the time may be very long

24. Why is path smoothing often applied after sampling-based planning?

- A) To add more collisions to the route
- B) To convert the route into a global map
- C) To increase unnecessary turns
- D) To reduce path length and produce a smoother, easier-to-track path

ANS: D) To reduce path length and produce a smoother, easier-to-track path

25. What is the function of speed-profile generation?

- A) Assigning a speed value to every point along a path
- B) Selecting the destination city

- C) Classifying road signs
 - D) Building the occupancy grid
- ANS: A) Assigning a speed value to every point along a path

26. What characterizes a linear-ramp speed profile?

- A) Constant speed at all times
- B) Constant acceleration until the reference speed is reached
- C) Random acceleration and braking
- D) A mandatory full stop at every path point

ANS: B) Constant acceleration until the reference speed is reached

27. Which profile is described as constant-zero-constant deceleration and can be used to stop at a red light?

- A) A circular profile
 - B) A linear-ramp profile
 - C) A trapezoidal profile
 - D) A probabilistic roadmap
- ANS: C) A trapezoidal profile

28. Why is 'avoid collisions at all costs' not a practical autonomous-driving objective?

- A) It requires all vehicles to drive faster
 - B) It eliminates the need for safety rules
 - C) It guarantees the AV is responsible for every crash
 - D) It can make the system so conservative that it becomes useless
- ANS: D) It can make the system so conservative that it becomes useless

29. What is the central safety goal of Responsibility-Sensitive Safety (RSS)?

- A) The AV should not cause accidents and should properly respond to others' mistakes
- B) The AV should never be involved in any collision under any circumstances
- C) The AV should always claim right-of-way
- D) The AV should ignore implicit traffic rules

ANS: A) The AV should not cause accidents and should properly respond to others' mistakes

30. In RSS, what is 'blame time'?

- A) The moment an insurance claim is filed
 - B) The first moment when a situation becomes dangerous
 - C) The time at which route planning begins
 - D) The delay between two sensor frames
- ANS: B) The first moment when a situation becomes dangerous